

The Average Eccentricity of a Graph and its Subgraphs

Peter Dankelmann, Wayne Goddard, Christine S. Swart
University of Natal, Durban

ABSTRACT. The eccentricity of a vertex is the maximum distance from it to another vertex and the average eccentricity of a graph is the mean eccentricity of a vertex. In this paper we establish bounds on the mean eccentricity of a graph. We then examine the change in the average eccentricity when a graph is replaced by a spanning subgraph, in particular the two extreme cases: taking a spanning tree and removing one edge.

1 Introduction

The eccentricity of a vertex is the maximum distance from it to any other vertex. Thus for example, the radius of the graph is the minimum eccentricity, and the diameter the maximum eccentricity. Nestled in between is the average eccentricity; this was introduced by Buckley and Harary [1] (as the eccentric mean), but seems not to have been studied.

Consider a communications network modelled by a graph with vertices representing the nodes of the network and edges representing the links between them. One might want to minimise the average, taken over all the nodes in the system, of the maximum time delay of a message emanating from it. This is the average eccentricity of the corresponding graph.

Often one considers the removal of links from a network. We are interested in two particular questions. The first is: how much damage can the failure of a link cause? That is, what is the maximum possible increase in a parameter when one edge is removed? Obviously, if the edge is a bridge then the graph will be disconnected, so we consider only the removal of edges that lie in cycles. We will refer to the cyclic edge maximising this increase as the “worst” edge; the “best edge” is one whose removal has least impact.

The second question is one of network design. Often the cheapest possible network is a tree. We investigate how much the parameter differs from that of a “best” spanning tree (i.e., one in which the parameter is

minimised). One could also investigate the spanning tree where the parameter is maximised, but this is related to the question of the existence of a hamilton path and we do not study it here.

In this paper we give results for the average eccentricity. We start with bounds, in particular we show that a graph with order n and minimum degree δ has average eccentricity at most $\frac{9}{4}n/(\delta+1)+O(1)$. For the network questions described above, we show that the best spanning tree has average eccentricity at most twice that of the original graph, and that the removal of any edge can increase the average eccentricity by at most a factor of 3.

1.1 Definitions and notation

We denote the vertex set of a graph G by $V(G)$, and the edge set by $E(G)$. The order of G is $n = |V(G)|$, and the size is $|E(G)|$. For $S \subseteq V(G)$, $\langle S \rangle_G$ denotes the subgraph induced in G by S .

The **distance** $d_G(u, v)$ between two vertices u and v is the number of edges in a shortest u - v path in G . The **eccentricity** $e_G(v)$ of v is the distance from v to a vertex farthest away from it in G . The **radius**, **diameter** and **average eccentricity** of G are, respectively, the minimum, maximum and mean eccentricity of a vertex in G . That is,

$$rad(G) = \min_{v \in V(G)} e_G(v), \quad diam(G) = \max_{v \in V(G)} e_G(v), \quad avec(G) = \frac{1}{n} \sum_{v \in V(G)} e_G(v).$$

Of course, $rad(G) \leq avec(G) \leq diam(G)$.

A vertex c of G is called **central** if $e_G(c) = rad(G)$. The **centre** $C(G)$ is the set of all central vertices in G . An **eccentric vertex** of a vertex v is a vertex farthest away from v .

If $S \subseteq V(G)$, then the **distance** $d_G(v, S)$ from a vertex v to S is the minimum distance from v to a vertex in S . The **distance** of S , denoted by $\sigma_G(S)$, is the sum of the distances from S to every vertex in G . In particular, the **distance** (or status) of a vertex v , is

$$\sigma_G(v) = \sum_{u \in V(G)} d_G(v, u).$$

Further, the **average distance** of a graph G is

$$\mu(G) = \frac{1}{n(n-1)} \sum_{v \in V(G)} \sigma_G(v)$$

If ϕ is a parameter, then a spanning tree T of graph G is **ϕ -preserving** if $\phi(T) = \phi(G)$. For example, T is **eccentricity-preserving** if $e_T(v) = e_G(v)$ for every vertex $v \in V(G)$, and **radius-preserving** if $rad(T) = rad(G)$.

2 Basic Results

Routine calculations yield the average eccentricity of the complete graph, the cycle, the path and the complete bipartite graph.

Theorem 1

- a) $avec(K_n) = 1$
- b) $avec(C_n) = \lfloor \frac{1}{2}n \rfloor$
- c) $avec(P_n) = \frac{1}{n} \lfloor \frac{3}{4}n^2 - \frac{1}{2}n \rfloor$ (which simplifies to $\frac{3}{4}n - \frac{1}{2}$ if n is even)
- d) $avec(K_{n_1, n_2}) = 2$ (if $n_1, n_2 \geq 2$).

It is easy to see that

$$avec(G) \geq \mu(G)$$

but there is no nontrivial lower bound on μ in terms of $avec$.

The following formula shows that the average eccentricity of a tree is determined by the eccentricity and distance of the centre.

Theorem 2 For any graph G ,

$$avec(G) \leq \frac{1}{n} \sigma(C(G)) + rad(G),$$

with equality for any tree, *inter alia*.

PROOF. For any vertex u of G , a walk to any other vertex can be obtained by going via the nearest central vertex. Thus $e_G(u) \leq rad(G) + d_G(u, C(G))$. Hence by averaging over all u we obtain the above bound. If G is a tree, then $e_G(u) = rad(G) + d_G(u, C(G))$, so there is equality in the bound. QED

Since the path has maximum radius and its center has maximum distance, we get the following corollary.

Corollary 1 The connected graph with maximum average eccentricity for given order is the path.

For integers $d, a, b \geq 1$, we define the **double broom** $D(d, a, b)$ as the tree obtained from the path P_{d-1} by attaching a endvertices to one end and b endvertices to the other. The double broom has diameter d , order $n = d + a + b - 1$ and for even diameter has average eccentricity $d - d^2/(4n)$.

Corollary 2 The graph with maximum average eccentricity for given order n and radius r is any double broom of diameter $2r$.

(These graphs maximise the value of $\sigma(C(G))$, since it is well known that for $i \in \{1, \dots, r-1\}$ there must be at least two vertices at distance i from the centre.)

In contrast, if we specify the order n and diameter d , then the maximum average eccentricity is d provided $n \geq 2d$, with extremal graph a cycle or a cycle with vertices duplicated.

Since the diameters of a graph and its complement are closely intertwined, the Nordhaus–Gaddum results for average eccentricity are easy to obtain.

Theorem 3 *Assume both G and its complement $\bar{G}$ are connected and $n \geq 5$. Then $4 \leq \text{avec}(G) + \text{avec}(\bar{G}) \leq 2 + \text{avec}(P_n)$ and $4 \leq \text{avec}(G)\text{avec}(\bar{G}) \leq 2\text{avec}(P_n)$, and these results are best possible.*

PROOF. Since both G and its complement are connected, each has radius at least 2. (In particular the lower bounds are trivial.) If one graph has diameter 2, then the other can have maximum average eccentricity, namely $\text{avec}(P_n)$. If both have diameter more than 2, then both have diameter equal to 3, and hence average eccentricity is at most this value. Since $\text{avec}(P_7) > 9/2$ and $\text{avec}(P_6) = 4$, the upper bound on the product holds for $n \geq 7$ and the upper bound on the sum holds for $n \geq 6$. A tedious calculation verifies the bounds in the remaining cases. QED

It is to be noted that the path is the (only and hence) extremal graph for $n = 4$ also.

Theorem 4 *If G is k -connected, then $\text{avec}(G) \leq n/k$ and this is best possible (at least for k even).*

For equality take expansion of the cycle: Let k be even and n a multiple of $k/2$. Take a cycle on $2n/k$ vertices and replace each vertex by a set of $k/2$ vertices with the same neighbourhood.

3 Bound Given Order and Minimum Degree

For an upper bound on the average eccentricity given the order and minimum degree we use the approach given in [3]. We consider weighted graphs where the vertices are weighted and the weights are nonnegative real numbers. For a weighted graph G with weight function $c : V(G) \rightarrow \mathbb{R}$, we define the eccentricity of G with respect to c by

$$EX_c(G) = \sum_{x \in V(G)} c(x)e_G(x)$$

and the average eccentricity of G with respect to c by

$$avec_c(G) = \frac{EX_c(G)}{\sum_{x \in V(G)} c(x)}.$$

We will need the following lemma which shows that (as expected) the path is extremal.

Lemma 1 *Let G be a weighted graph with weight function c such that $c(v) \geq 1$ for every vertex v . Then for $N = \lceil \sum_{v \in V(G)} c(v) \rceil$,*

$$avec_c(G) \leq avec(P_N).$$

PROOF. It suffices to prove the result for trees T . We prove the equivalent statement $EX_c(T) \leq EX(P_N)$.

Let T be a tree with weight function c satisfying the hypothesis of the lemma such that $EX_c(T)$ is maximum. We may assume that the total weight of the vertices in T equals N . We show that T is a path.

Suppose not. Let $P = v_1, v_2, \dots, v_p$ be a longest path of T and let v_i be a vertex on P of degree at least three. For each vertex of T , at least one of the vertices v_1 and v_p is an eccentric vertex. Let u be a neighbour of v_i in T which is not on P . Without loss of generality we can assume that v_1 is an eccentric vertex of u and thus of every vertex of the component T_u of $T - V(P)$ containing u .

Consider the graph $T' = T - uv_i + uv_p$. Then for each vertex in T_u , the distance to v_1 and thus its eccentricity has increased by $p - i$. On the other hand, for every vertex not in T_u , the distances to v_1 and v_p have remained unchanged, hence its eccentricity has not decreased. Therefore, $EX_c(T') \geq EX_c(T) + (p - i)|V(T_u)|$, contradicting the choice of T . Hence T is a path $v_1, v_2, \dots, v_n$.

Since the only vertices with eccentricity $n - 1$ are the end-vertices of T , it follows that all other vertices must have weight as small as possible, viz. 1. Finally, we show that $n = N$. Suppose not. Then some end-vertex, say v_n , has weight at least 2. If we add a vertex v_{n+1} adjacent to v_n and transfer 1 unit of weight to it, we obtain a weighted tree with higher eccentricity, a contradiction. Hence $T = P_N$. QED

By a **packing** we mean a set of vertices such that the distance between any pair of vertices is at least 3.

Theorem 5 *Let G be a connected graph with n vertices and minimum degree δ . Then*

$$avec(G) \leq \frac{9n}{4\delta + 4} + \frac{15}{4}. \quad (1)$$

Apart from the additive constant, this inequality is best possible.

PROOF. First we find a maximal packing A of G as follows. Choose an arbitrary vertex v of G and let $A = \{v\}$. If there exists a vertex u in G with $d_G(u, A) = 3$, add u to A . Add vertices u with $d_G(u, A) = 3$ to A until every vertex not in A is within distance two of A .

Let T_1 be the forest subgraph with vertex set $N[A]$ (that is, A and any vertex adjacent to A) and whose edge set consists of all edges incident with a vertex in A . By the construction of A , there exist $|A| - 1$ edges in G , each one joining two neighbours of distinct elements of A , whose addition to T_1 yields a subtree T_2 of G . Now, each vertex $u \in V(G) - N[A]$ is adjacent to some vertex $u' \in N[A]$. Let T be the spanning tree of G with edge set $E(T_2) \cup \{uu' : u \in V(G) - N[A]\}$.

It suffices to prove that

$$avec(T) \leq \frac{9n}{4\delta + 4} + \frac{15}{4}.$$

For every vertex $u \in V(T)$, let u_A be a vertex in A closest to u in T . We move the weight of u to u_A . That is, we define a weight function $c : V(T) \rightarrow \mathbb{R}$ by

$$c(u) = |\{x \in V(T) \mid x_A = u\}| \quad \text{for } u \in V(T).$$

Note that $\sum_u c(u) = n$, $c(u) = 0$ if $u \notin A$ and

$$c(u) \geq \delta + 1 \quad \text{for each } u \in A.$$

Since each weight was moved over a distance not exceeding two,

$$avec(T) \leq avec_c(T) + 2.$$

Now, define the tree U with vertex set A by taking T , and deleting all non- A vertices that are end-vertices, and contracting out all non- A vertices of degree 2. Since only the vertices of A have positive weight, it follows that

$$avec_c(T) \leq 3avec_c(U) + 1.$$

Let $N = \lceil n/(\delta + 1) \rceil$. By Lemma 1, with $c'(u) = c(u)/(\delta + 1)$,

$$avec_{c'}(U) \leq avec(P_N) \leq \frac{3N}{4} - \frac{1}{2}.$$

Combining these inequalities, in conjunction with $N < n/(\delta + 1) + 1$, yields

$$avec(T) \leq \frac{9N}{4} - \frac{3}{2} + 3 < \frac{9n}{4\delta + 4} + \frac{15}{4},$$

as desired.

It remains to show that, apart from the value of the additive constant, the bound (1) is best possible. For given integers n, δ, k with $n = k(\delta+1)$, let $G_1, G_2, \dots, G_k$ be disjoint copies of the complete graph $K_{\delta+1}$ and let $a_i b_i \in E(G_i)$. Let $G_{n,\delta}$ be the graph obtained from the union of $G_1, G_2, \dots, G_k$ by deleting the edges $a_i b_i$ for $i = 2, 3, \dots, k-1$ and adding the edges $a_{i+1} b_i$ for $i = 1, 2, \dots, k-1$. It is easy to verify that for large n ,

$$\text{avec}(G_{n,\delta}) = \frac{9n}{4\delta + 4} + O(1).$$

This completes the proof. QED

4 Spanning Trees

For radius, it is well known that every connected graph has a radius-preserving spanning tree (which can be obtained by a breadth-first search from any central vertex). For diameter, the question was answered by Hakimi et al. [4].

Theorem 6 *Let T be a minimum-diameter spanning tree of a graph G . If G contains a pair of adjacent central vertices with no eccentric vertices in common, then $\text{diam}(T) = 2\text{rad}(G) - 1$. Otherwise, $\text{diam}(T) = 2\text{rad}(G)$.*

Since taking a spanning tree cannot decrease the eccentricity of any vertex, a spanning tree is *avec*-preserving if and only if it is eccentricity-preserving. Graphs which have *avec*-preserving spanning trees were characterised by Nandakumar and Parthasarathy [5].

Theorem 7 [5] *A connected graph G has an eccentricity-preserving spanning tree iff*

- (1) *either $\text{diam}(G) = 2\text{rad}(G)$ and G has a unique central vertex, or $\text{diam}(G) = 2\text{rad}(G) - 1$ and G has two central vertices and these two are adjacent; and*
- (2) *every non-central vertex has a neighbour with smaller eccentricity.*

It is easy to show that the average eccentricity of the best spanning tree is at most double that of the original graph. For, if we take a distance-preserving spanning tree T_v rooted at a vertex v , then by the same argument as used in the proof of Theorem 2, $\text{avec}(T_v) \leq e(v) + \sigma(v)/n$. If we average the latter quantity, it follows that the best distance-preserving tree T has

$$\text{avec}(T) \leq \text{avec}(G) + \mu(G).$$

The next result characterises how closely the average eccentricity of a best spanning tree of a graph G can approach $2\text{avec}(G)$.

Theorem 8 *Let T be a minimum-avec spanning tree of a graph G of order n . Then*

$$\text{avec}(T) \leq \text{avec}(G) \left(2 - \frac{1}{n} \text{avec}(G)\right),$$

with equality iff G is complete or an odd cycle.

PROOF. Let c be a central vertex of G , and let T_c be a spanning tree of G which is distance-preserving from c . Then $\text{rad}(T_c) = \text{rad}(G) = r$. By Theorem 2 and Corollary 1,

$$\text{avec}(T) \leq \text{avec}(T_c) \leq \text{avec}(D_{2r,x,y}) = 2r - r^2/n \leq \text{avec}(G) \left(2 - \frac{1}{n} \text{avec}(G)\right).$$

For the bound to be attained, T_c must be a star or double broom, and hence c must have degree 2 or $n - 1$. Furthermore, $C(G)$ must be $V(G)$. Since c could be *any* central vertex of G , this means that every vertex in G must have degree 2 or $n - 1$. It follows that equality holds iff G is a complete graph or an odd cycle. QED

Note that a minimum-avec tree need not be radius-preserving. For example, take a cycle $C : v_0, \dots, v_{2r-1}, v_0$, add k endvertices adjacent to v_1 , and join one endvertex to each other vertex of C except v_0 . Then v_r is the only central vertex of G , and the only radius-preserving spanning trees of G are distance-preserving from v_r and omit an edge incident with v_0 . If k is large enough, then $\text{avec}(T_{v_r}) \geq 2r - 2$, while $\text{avec}(T_{v_1}) \leq r + 3$ —so the minimum-avec spanning tree is not radius-preserving.

For specific families of graphs, one can exhibit much stronger bounds. For example, Prisner [8] showed that in a chordal graph one can always find a spanning tree T such that for every vertex v

$$e_T(v) \leq e_G(v) + 2.$$

We do not know the complexity of finding a minimum-avec spanning tree.

5 Edge Removal

Removing an edge from a graph cannot decrease the radius; however, every connected cyclic graph has a radius-preserving spanning tree, and hence contains at least one edge that can be removed without increasing the radius. Segawa [9] showed that edge removal in connected graph can increase the radius by at most a factor of 2.

For diameter the results are similar. The removal of a cyclic edge can at most double the diameter. This was first observed Plesník [6, 7], who went on to show that if the graph has minimum degree at least 2, then the diameter doubles on the removal of an edge iff the shortest cycle containing that edge has length $2\text{diam}(G) + 1$. Another class of graphs which have been studied are the **diameter-minimal** graphs: graphs for which the removal of any edge increases the diameter.

Turning to average eccentricity, the following is probably known, but we do not know a reference. If e is an edge of G , then $d_G(v, e)$ denotes the shortest distance between v and a vertex incident with e .

Lemma 2 *Let v be a vertex of a graph G , and let $e = ab$ be a cyclic edge of G . If a and b are both eccentric vertices of v , then $e_{G-e}(v) = e_G(v)$. Otherwise*

$$e_{G-e}(v) \leq 3e_G(v) - 2d_G(v, e) - 1.$$

PROOF. Consider any tree T that is distance-preserving from v . If e is not in the tree T then $e_{G-e}(v) = e_G(v)$, and the result follows. So assume T does contain e , and without loss of generality let $d_T(v, a) < d_T(v, b)$. Let T_a and T_b denote the components of $T - e$ containing a and b respectively.

Let x be any vertex. If x is in T_a then $d_{G-e}(v, x) \leq e_G(v)$. So assume x is in T_b . Note that $\text{rad}(T_b) \leq e_{T_b}(b) \leq e_G(v) - d_G(v, e) - 1$. Hence T_b has diameter at most $2(e_G(v) - d_T(v, a) - 1)$. Since e is a cyclic edge, there is some edge joining a vertex of T_a , say u , to a vertex of T_b , say w . It follows that

$$d_{G-e}(v, x) \leq d_{T_a}(v, u) + 1 + d_{T_b}(w, x) \leq e_G(v) + 1 + 2(e_G(v) - d_G(v, a) - 1).$$

This completes the proof. QED

The bound is best possible: take a cycle of length $2r + 1$ containing an edge ab and attach a path of length $r - 1$ to b (so that $e_G(a) = r$ and $e_{G-ab}(a) = 3r - 1$).

A consequence of the above theorem is the following result.

Theorem 9 *If e is a cyclic edge of a graph G of order n and radius at least 2, then*

$$\text{avec}(G) \leq \text{avec}(G - e) \leq 3\text{avec}(G) - 3 + 4/n.$$

PROOF. By Lemma 2, $e_{G-e}(v) \leq 3e_G(v) - 1$ if v is incident with e , and $e_{G-e}(v) \leq 3e_G(v) - 3$ otherwise. The bound follows. QED

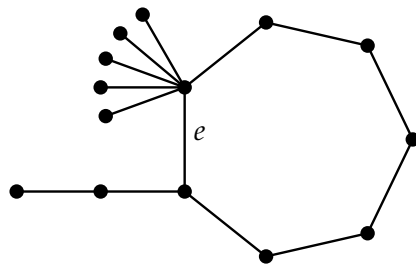


Figure 1: A graph G with $avec(G - e) \approx 3avec(G) - 3$

The following example shows that the bound cannot be much improved for general graphs. Let G be the graph obtained from a cycle of length $2r + 1$, containing an edge $e = ab$, by attaching a path of length $r - 1$ to b and k endvertices to a . If k is large, the average eccentricity of G is $r + 1 + o(1)$, while that of $G - e$ is $3r + o(1)$. So $avec(G - e)$ is approximately equal to $3avec(G) - 3$. (See Figure 1.)

We note in passing that all diameter-minimal graphs are also avec-minimal, in the sense that removing any edge increases the average eccentricity. By contrast, there are many graphs where the removal of any edge leaves the eccentricity of every vertex (and hence the average eccentricity) the same; for example, the cartesian product $G \times H$, where G and H are any connected graphs of radius at least 2.

When it comes to the removal of the best edge, we believe the following to be true.

Conjecture 1 *For every cyclic connected graph G there exists a cyclic edge e such that*

$$\frac{avec(G - e)}{avec(G)} \leq \frac{3}{2} + o(1).$$

This bound is best possible, as it is attained, for example, by the cycles.

As for adding edges, Torbett et al. [11] characterised those edges which can be added to a tree without affecting the eccentricity of any vertex.

References

- [1] F. Buckley and F. Harary, *Distance in Graphs*, Addison-Wesley, Redwood City, California (1990).
- [2] F. Buckley and M. Lewinter, A note on graphs with diameter-preserving spanning trees, *J. Graph Theory* **12** (1988), 525–528.

- [3] P. Dankelmann and R. Entringer, Average distance, minimum distance, and spanning trees, *J. Graph Theory* **33** (2000), 1–13.
- [4] S.L. Hakimi and E.F. Schmeichel, On p -centers in networks, *Transportation Sci.* **12** (1978), 1–15.
- [5] R. Nandakumar and K.R. Parthasarathy. Eccentricity-preserving spanning trees *J. Math. Phys Sci* **24** (1990), 33–36.
- [6] J. Plesník, Critical graphs of given diameter, *Acta Fac. Rerum Natur. Univ. Comenian. Math.* **30** (1975), 71–93.
- [7] J. Plesník, Note on diametrically critical graphs, in: Recent Advances in Graph Theory (Proc. Second Czechoslovak Sympos. Prague 1974), Academia, Prague (1975) 455–465.
- [8] E. Prisner, Eccentricity-approximating trees in chordal graphs, *Discrete Math.* **220** (2000), 263–269.
- [9] T. Segawa, Radius increase caused by edge deletion, *SUT J. Math.* **30** (1994), 159–162.
- [10] C.S. Swart, Distance in Graphs and Subgraphs, M.Sc. thesis, University of Natal, Durban, 1997.
- [11] P.O. Torbett, J.W. Boland and L. Lawson, Eccentricity stable edges of trees, *Congr. Numer.* **118** (1996), 203–208.